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LARSON—MATH 756—HOMEWORK WORKSHEET 06
Kőnig-Egerváry graphs, Forbidden subgraphs & Good Characterizations

A *Kőnig-Egerváry graph* is a graph where the independence number α plus the matching number μ equals its order n . Call them *KE graphs*. It turns out that these graphs can be identified efficiently—and, as the matching number can be computed efficiently, this is a class of graph where the independence number can be computed efficiently.

The Kőnig-Egerváry Theorem implies that all bipartite graphs are KE graphs. (The paw graph is an example of a non-bipartite KE graph.)

1. Let G be a graph. Argue that G is an induced subgraph of *some* KE graph G' .

Let P be a graph property (so there is a set of graphs that have this property; being *bipartite*, for instance, is a graph property). A *good characterization* of the graphs with property P is given by identifying a graph property Q such that an arbitrary graph has exactly one of property P or property Q , and an efficiently checkable *certificate* can be provided that proves the graph has one of the properties. So,

A graph has property P if and only if it does **not** have property Q .

An example is if property P is “being bipartite” and property Q is “having an odd cycle”. We know:

A graph is bipartite if and only if it does **not** have an odd cycle .

If a graph is bipartite, we can demonstrate or prove that by exhibiting a bipartition (we can efficiently check that this *is* a bipartition. If a graph has an odd cycle we can demonstrate that by exhibiting the cycle (we can efficiently check that a provided list of vertices *is* a cycle and that it contains an odd number of vertices).

It is also worth noting that this is a *forbidden subgraph* characterization: a graph is bipartite if and only if it does not have a subgraph of a certain (forbidden) type.

Lovász and Plummer (1984) provide a forbidden subgraph characterization of KE graphs. Their characterization is restricted to graphs with a matching that saturates all vertices (a *perfect matching*, where $\mu = \frac{n}{2}$).

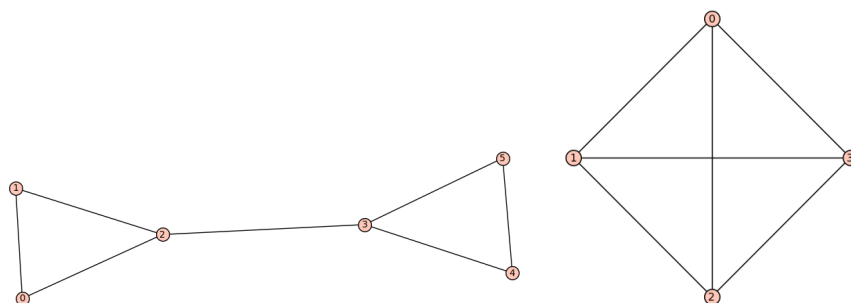
We first prove that, for the purposes of investigating the independence number α , it is not a limitation to restrict one’s attention to graphs with a perfect matching.

2. Suppose G is a graph with matching number $\mu(G) < \frac{n(G)}{2}$ and maximum matching M . Let V be the vertices of G , $V(M)$ be the vertices saturated by M and $U = V \setminus V(M)$ be the vertices not saturated by M . Let $U = \{u_1, \dots, u_k\}$.

We will construct a new graph G' , with induced subgraph G , having a perfect matching, and with $\alpha(G') = \alpha(G)$. For each vertex $u_i \in U$ we will create an adjacent twin (so having the same neighbors as u_i). Let $U' = \{u'_1, \dots, u'_k\}$. For $i \in \{1, \dots, k\}$, let $u_i \tilde{u}_k$ and, for every neighbor w of u_i in G , let w be adjacent to u'_i in G' .

- (a) Get G be K_3 . Find a maximum matching M and the corresponding G' . Check that G' has a perfect matching and $\alpha(G') = \alpha(G)$.
- (b) Let G be your own choice of a graph without a perfect matching. Find a maximum matching M and the corresponding G' . Check that G' has a perfect matching and $\alpha(G') = \alpha(G)$.
- (c) Prove: if G is a graph without a perfect matching, M is a maximum matching and G' is the corresponding extension then (1) G' has a perfect matching and (2) $\alpha(G') = \alpha(G)$.
- (d) Let G be a graph with maximum matching M and corresponding extension G' . Prove: G is KE if and only if G' is KE.

The Lovász-Plummer characterization of KE graphs involves even subdivisions of the following graphs, T and K_4 .



3. Show: that the T graph (left) is not KE.
4. Show: that the K_4 graph (right) is not KE.
5. Show: that no even subdivision of the T graph is KE.
6. Show: that no even subdivision of the K_4 graph is KE.

An *even subdivision* of a graph is the (possibly iterated) process of adding two new vertices to any existing edge.

A subgraph H of a graph G is *nice* if the subgraph induced by the vertices not in H has a perfect matching.

7. Find a graph G that has T as a nice subgraph.
8. Find a graph G that has K_4 as a nice subgraph.

Lovász and Plummer prove: **If G is a graph with a perfect matching then G is KE if and only if G does not have an even subdivision of either of T or K_4 as a nice subgraph.** This is a forbidden subgraph characterization **and** a good characterization.

So, given a graph G with a perfect matching, the theorem says that exactly one is true: either G is KE or it **does** have an even subdivision of T or K_4 as a nice subgraph. This condition can be checked efficiently: produce the even subdivision of T or K_4 and then show the graph induced on the remaining vertices has a perfect matching.

9. Suppose G is a graph with a perfect matching M . How can you demonstrate or prove that G is KE (in a way that can be efficiently checked)?